Data Sources

Inputs, transformations and access conditions for the completed VaR/ES study

This record describes the data actually used in the frozen evaluations and the development data that informed their fixed settings. It is based on the retained downloads, loaders, execution receipts and final manuscript. No new dataset was substituted, and no empirical result was changed.

1. US industry portfolios: Study 1

Provider. Kenneth R. French Data Library, Tuck School of Business at Dartmouth. The portfolio series are constructed from CRSP records; the project did not obtain the underlying individual-security CRSP database.

Acquisition. Data Library; 30-industry construction notes; daily CSV ZIP download. The retained file was downloaded on 23 October 2026 at 22:23:33 UTC. Its header identifies the 202608 CRSP database.

Item	Actual input or operation
Section used	Average Value Weighted Returns -- Daily
Study interval	1 July 1963-31 August 2026; 15,897 daily returns per industry
Evaluation interval	3 July 1975-31 August 2026; 12,897 common dates per industry
Variables	Date and 30 industry return columns, in percentage simple returns
Transformation	y[t] = 100 * log1p(r[t] / 100)
Cleaning	Parse YYYYMMDD; restrict to dates from 1963-07-01; remove -99.99, -999 and nonfinite values separately by series. No removals occurred in the retained study interval.

Industry columns retained

Food, Beer, Smoke, Games, Books, Hshld, Clths, Hlth, Chems, Txtls, Cnstr, Steel, FabPr, ElcEq, Autos, Carry, Mines, Coal, Oil, Util, Telcm, Servs, BusEq, Paper, Trans, Whlsl, Rtail, Meals, Fin, Other.

The equal-weighted section, risk-free rate, factor series, firm counts and firm-size sections are not used. There is no imputation, winsorization or additional return adjustment.

Distribution

The provider makes the download public but displays a Fama/French copyright notice. An explicit redistribution grant was not established. The distribution ZIP therefore omits the raw archive and reversible return copies. The exact URL, expected bytes and SHA-256 are retained; authorized original files are required for an exact Study 1 replay.

International Equity Indices

Study 1 secondary index sample

The provider actually accessed was Yahoo Finance. The six archived JSON responses supply timestamp, meta.gmtoffset and indicators.quote[0].close. The request included adjusted-close data, but the estimator uses the close field. Open, high, low and volume are not predictors. Yahoo symbols have a leading caret (for example, ^TWII); the tables omit this prefix.

Symbol	Return sample	Returns	Evaluation start / N
TWII	1997-07-03 to 2026-10-22	7181	2009-09-14 / 4181
KS11	1996-12-12 to 2026-10-22	7348	2009-02-24 / 4348
FTSE	1984-01-04 to 2026-10-23	10814	1995-11-16 / 7814
GDAXI	1988-01-04 to 2026-10-22	9815	1999-12-21 / 6815
AXJO	1992-11-24 to 2026-10-23	8575	2004-09-30 / 5575
GSPTSE	1979-07-03 to 2026-10-23	11880	1991-06-25 / 8880

Acquisition and identity

The frozen requests used Yahoo Finance chart/v8 with daily frequency, period1=0 and period2=1792794213. Retrieval occurred on 23 October 2026 at 22:23:33-22:23:40 UTC. Exact URLs and all six hashes are in data/study1/expected_fetch_manifest.json.

The symbols represent Taiwan Weighted, KOSPI, FTSE 100, DAX, S&P/ASX 200 and S&P/TSX Composite. GDAXI is labelled "DAX P" in the response; it is not relabelled as a total-return series.

Cleaning and return construction

Dates are formed from the saved Unix timestamps plus the saved GMT offset. The last record is kept for a duplicate date. Nonfinite and nonpositive closes are removed; returns are 100 * log(P[t]/P[t-1]) across consecutive retained levels. Missing market dates are not filled. Historical-timezone conversion was separately checked against the archived dates and produced no date differences.

Series	Raw rows	Duplicates	Invalid closes	Retained levels
TWII	7324	0	142	7182
KS11	7499	0	150	7349
FTSE	10998	0	183	10815
GDAXI	9985	0	169	9816
AXJO	8678	1	101	8576
GSPTSE	12131	0	250	11881

The six evaluation calendars have a union of 9,119 dates, 25 June 1991-23 October 2026. Panel scores average over series observed on each date, preserving the original unequal calendar coverage.

Access conditions

Raw redistribution permission was not established. Yahoo download guidance identifies subscription and licensing restrictions; Yahoo terms do not establish unrestricted automated access. The old endpoint is a provenance record. An exact replay requires authorized original inputs; a later download may differ.

Historical Yen and Development Data

2. Historical yen: Study 2

Originating source: Board of Governors of the Federal Reserve System, H.10. **Distributor:** FRED, Federal Reserve Bank of St. Louis. DEXJPUS is Japanese yen per US dollar, daily and not seasonally adjusted. Negative log returns indicate yen appreciation.

The historical CSV request covers 2 January 1974-31 December 2009. The original response, downloaded 24 October 2026 at 01:00:07.647972 UTC, is included as data/study2/DEXJPUS_1974_2009.csv.

The loader parses dates and numeric levels, restricts the requested interval, removes 348 missing levels, and requires unique increasing dates and positive levels. It retains 9,044 levels and 9,043 returns. Returns are 100 * log(P[t]/P[t-1]); the quotation is not inverted. There is no interpolation, forward fill or post-2009 input.

Stage	First / last relevant date
First return	3 January 1974
First 1,000 returns end	29 December 1977
Score-history accumulation starts	30 December 1977
Calibration starts	29 December 1981
Evaluation	24 December 1985-31 December 2009; 6,043 returns

DEXJPUS is tagged "Public Domain: Citation Requested." The Federal Reserve Board copyright policy permits copying Board information unless an exception applies and requests attribution. These source-specific conditions support inclusion of this CSV; they do not imply that every FRED series is redistributable.

3. Development data: parameter provenance only

The final development rounds used five FRED series. They informed method design; SP500, DJIA and NASDAQCOM score differences determine the stored bootstrap blocks. These are not additional confirmation observations and are not required to execute the frozen evaluations.

FRED series	Observed daily level interval	Originating provider
SP500, DJIA	2016-10-24 to 2026-10-21	S&P Dow Jones Indices LLC
NASDAQCOM	2016-01-04 to 2026-10-21	Nasdaq, Inc.
DEXUSEU, DEXJPUS	2010-01-04 to 2026-10-16	Federal Reserve Board H.10

Development returns use the same consecutive-valid-level log transformation. DEXUSEU is US dollars per euro; DEXJPUS is yen per dollar. Exact requests, counts, download times and matching source hashes are in provenance/Development_Provenance.md. FRED notes reserve redistribution rights for the three equity series. Development FX raw data are unnecessary for the fixed-parameter replay and are omitted.

Current development score files reproduce all six frozen block lengths. Those score files lack original freeze-date hash receipts and were absent from the handoff archive; this consistency check does not establish their historical byte identity. The retained freeze_params.json is never overwritten.

Input Identity and Reproduction Boundaries

All eight evaluation input files were hashed locally against the original acquisition manifests. Matching a URL alone does not establish the same data vintage. The raw Study 1 inputs remain external to the distributed package.

Input	SHA-256
30_Industry_Portfolios_daily_CSV.zip	04a7a74fcd7c12e90461718e5ba8490e99d18c3e9323182b6c7a7b320e6cd34a
yahoo_TWII.json	0ca89944534f38a8b298c0899a73df02fb37953895797a6c7ae80f1ab8b93329
yahoo_KS11.json	03d456451d3fa3b068044c8d29b9a1046db7c2f5152f8a8d2accfd57b5167b0f
yahoo_FTSE.json	a8e62f7fb39e9c68fbe5c9e96f638c8aa8461fdf90ebff436dadb183a6ae0edf
yahoo_GDAXI.json	bb64d171189e255c12748f55c34f1894b70b5e0426327bb29b44b166b5d7ce60
yahoo_AXJO.json	20921f149b40432bcb20baa4fd401bd339a6a375565a27b56664c2ed1c6b7e10
yahoo_GSPTSE.json	704065bd537bedd65a0a4fe60bce6bf81b3923d6bf86e30b70f7dcfbae52b519
DEXJPUS_1974_2009.csv	90fa5f9af7dd2ba1754958b11114e41f48946efa440f8afc9f62b50a09a479b3

Reproduction procedure

The README gives environment installation, fresh-workspace creation, input verification, model replay, bootstrap inference, table/figure reconstruction and numerical comparison commands. code/data/manage_inputs.py verifies exact identities and imports only matching lawfully obtained inputs. It refuses conflicting files. The historical downloader is retained as source evidence; it is not required for an offline replay.

Variables produced by the analysis

The principal derived variables are percentage log return y; lower-tail VaR q and ES e; the VaR violation indicator 1[y<q]; FZ0 score; the ES calibration diagnostic Z2; model/expert weights; forecast magnitude and change; unconditional-coverage and independence p-values; and method score differences. These are computed outputs, not additional external data sources.

Timing and scope

Forecast estimation, model selection and calibration use prior observations in the inspected code. The package additionally tests future-return perturbations. The 60-day yen-appreciation state includes the current return and a full-evaluation-sample threshold; signal tertiles and other risk-state boundaries also use full-sample cutoffs. They are retrospective descriptions and do not feed the forecasts.

Study 2 applies later-developed fixed settings to an earlier yen sample: it is a backward-in-time validation, not a prospective historical deployment. Study 1 holds out series, while economic histories overlap development in calendar time.

All historical series are the retained October 2026 vintages. Exact numerical reproduction does not establish point-in-time availability, freedom from provider revisions, constituent-level survivorship properties or the order of events before the protocol freezes, which rests on commit records kept by the author. No source or observation outside the actual final studies and directly documented development inputs has been added.

Provider access statements were checked on 25 October 2026. The package grants no new rights in third-party data.